OrdinalFed: Federated Learning with Ordinal-Aware Aggregation for Diabetic Retinopathy Grading

***Abstract*—Diabetic Retinopathy (DR) is a leading cause of preventable blindness. Federated learning (FL) allows hospitals to train a shared DR grading model without exchanging retinal images, but when hospitals see very different patients, standard FedAvg can collapse. OrdinalFed is studied as a remedy with two components: (1) an *OrdinalLoss* that penalises grading errors by their squared distance, and (2) a *QWK-weighted aggregation* in which the server scores each client model by its Quadratic Weighted Kappa (QWK) on a 50-image server set and moves the aggregation weights from image counts towards these scores over the rounds. On the DDR benchmark, every configuration was repeated with three seeds and scored on the held-out test split. Under extreme label skew (Dirichlet $\alpha = 0.1$), FedAvg collapsed in every seed (worst-round test QWK 0.378), and OrdinalFed raised the worst round to 0.514 without any change at the hospitals and with about 1.2 s of extra server computation per round. Ablations showed that this gain comes from giving small hospitals more weight: equal client weights did better (0.630), while the QWK scores and the OrdinalLoss added no measurable benefit. FedProx was the most stable method (0.641), and adding its proximal term to OrdinalFed came close to it (0.627). On the IID and realistic multi-site partitions, all methods were within 0.01 QWK of each other.**

***Index Terms*—Diabetic Retinopathy, Federated Learning, Non-IID, Ordinal Loss, QWK-Weighted Aggregation, EfficientNet-B3, Medical Imaging**

I. Introduction

Diabetic Retinopathy (DR) is a common complication of diabetes and a leading cause of preventable blindness in working-age adults. It damages the small blood vessels of the retina; in fundus photographs, the damage appears as microaneurysms, haemorrhages, hard exudates and, in advanced cases, the growth of new vessels. In countries such as Bangladesh, where more than 13.1 million people live with diabetes [1], the burden is made worse by a shortage of retinal specialists outside the major cities. DR is graded on a five-level ordinal scale: Grade 0 (no DR), Grade 1 (mild non-proliferative DR, NPDR), Grade 2 (moderate NPDR), Grade 3 (severe NPDR), and Grade 4 (proliferative DR). Referable disease (Grade ≥ 2) has to be detected early to prevent vision loss, which makes automated screening attractive.

When all images are pooled in one place, deep convolutional networks grade DR well. On the imbalanced DDR dataset [2], an EfficientNet-B3 model trained in this way reaches a Quadratic Weighted Kappa (QWK) of 0.834 on the validation split and 0.682 on the separate test split (Sec. V). QWK is the standard metric for ordinal grading because it penalises an error by the square of its distance: predicting Grade 0 for a Grade 4 image costs far more than predicting Grade 3. Pooling patient images from several hospitals, however, is restricted by privacy regulations such as HIPAA and GDPR. Federated Learning (FL) [3] avoids this problem: the hospitals train a shared model by exchanging model weights, and the images never leave each site.

The main difficulty in clinical FL is that hospitals see different patients, so their data are not identically distributed (non-IID). Screening clinics see mostly healthy (Grade 0) eyes, referral centres see many severe cases (Grades 3–4), and rural clinics see cases in between. Under extreme Dirichlet skew ($\alpha = 0.1$), standard FedAvg is observed to collapse: averaged over three seeds, its test QWK falls from 0.647 in its best round to 0.378 in its worst. FedAvg weights each hospital by its number of images, so a few large clinics that each see mostly one grade can dominate the global model. FedProx [4] avoids the collapse by limiting how far each local model can move away from the global model, but it changes the training at every hospital.

In addition, the standard cross-entropy loss ignores the order of the grades: mistaking Grade 4 for Grade 0 costs the same as mistaking it for Grade 3. Ordinal losses [5]–[6] address this in centralised training, but their effect on client drift in FL has received little attention.

To address both problems, **OrdinalFed** is proposed: a federated framework for ordinal medical grading with two components, together with a repeated evaluation that tests what each component contributes.

1) **Distance-aware OrdinalLoss:** a local training loss that adds a squared expected-grade penalty to class-weighted cross-entropy, so that predictions far from the true grade are penalised more heavily.
2) **QWK-weighted aggregation with an adaptive β:** the server scores each client model on a small balanced set $\mathcal{V}$ of 50 images and gradually shifts the aggregation weights from image counts towards these QWK scores, without any change at the hospitals.
3) **A multi-seed evaluation on held-out test data:** every configuration is run with three seeds and scored on the DDR test split, with ablations and a sensitivity analysis. It shows that OrdinalFed prevents most of FedAvg's collapse, that the gain comes from giving small hospitals more weight rather than from the QWK scores or the ordinal loss, and that FedProx remains the most stable method.

II. Related Work

Ordinal learning for DR. Standard multi-class losses treat severity grades as unrelated categories and ignore their order. Toledo-Cortés et al. [5] proposed Deep Quantum Ordinal

Regression (DQOR) and reported a grading mean absolute error (MAE) of 0.2871 on EyePACS. Polat et al. [6] proposed Class Distance Weighted Cross-Entropy (CDW-CE), which scales the loss by the distance between classes, and reached a QWK of 0.8568 in ulcerative colitis grading. Both studies show that distance-aware losses help in centralised training. To the best of the authors' knowledge, such losses have not yet been combined with quality-weighted aggregation in federated DR grading.

Federated learning in ophthalmology. FedAvg [3] is the standard FL algorithm. FedProx [4] adds an L_2 proximal term that keeps local weights close to the global model when client data differ. For DR, Bhulakshmi and Rajput [7] proposed FedDL with dynamic aggregation switching, and Ciaramella et al. [8] combined federated DR screening with Grad-CAM explanations across 20 non-IID clients on APTOS 2019. These frameworks treat grading as plain (nominal) classification and weight clients by dataset size rather than by the quality of their models.

Quality-aware aggregation. Weighting clients by sample count can fail when client data are strongly biased or adversarial. Bejenar et al. [9] showed that scoring client models on a server-side validation set (FedLasso) outperforms FedAvg by 45.6 pp under adversarial label flipping. Li et al. [10] showed that non-normalised client weights act as an implicit regulariser (FedLAW), and Li et al. [11] proposed FedAAW, which derives aggregation weights from gradient norms and improves non-IID accuracy by 37.32 pp. A review by Mora et al. [12] concluded that combining client-side regularisation with server-side adaptive aggregation gives the most robust results on non-IID data. OrdinalFed follows this direction by pairing an ordinal client loss with QWK-based server aggregation.

III. Methodology

A. Problem Formulation

$K = 5$ hospital clients each hold a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ of fundus images x_i with ordinal grades $y_i \in \{0, 1, 2, 3, 4\}$. A shared EfficientNet-B3 model with weights w is trained over T communication rounds, and no $\mathcal{D}_k$ ever leaves its client. Because the local grade distributions $P_k(y)$ differ strongly from the global distribution $P(y)$, the local models drift apart during local training.

B. Limitations of Nominal Objectives

FL clients are usually trained with cross-entropy (CE). With softmax probabilities $\mathbf{p}(x) \in \mathbb{R}^5$, the CE loss for true grade y is $-\log p_y(x)$. This loss does not distinguish a prediction of $y + 1$ from one of $y + 4$: both are simply wrong. On a heavily skewed client, for example a clinic with only Grade 0 patients, CE quickly pushes the local model to predict Grade 0 for every image, and the model loses what it had learned about severe disease.

C. Distance-Aware OrdinalLoss

To limit this degradation, a hybrid loss, OrdinalLoss, is used during local training:

$$\mathcal{L}_{\text{ord}}(f(x), y) = (1 - \lambda)\, \mathcal{L}_{\text{CE}}(f(x), y) + \lambda\, (\hat{y} - y)^2 \tag{1}$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy with fixed class weights $\mathbf{c} = [1.0, 5.0, 1.5, 10.0, 4.0]$ to counter class imbalance, and $\lambda = 0.5$ balances the two terms. $\hat{y}$ is the expected grade under the softmax distribution:

$$\hat{y} = \sum_{c=0}^{4} c \cdot p_c(x) \tag{2}$$

The penalty $(\hat{y} - y)^2$ grows with the square of the grade error: for a confident prediction, calling a Grade 4 case Grade 0 costs $16\times$ as much as calling it Grade 3. Local training is therefore discouraged from predictions far from the true grade, even when a client's data are dominated by a single grade.

D. Server Validation Set $\mathcal{V}$ and QWK Estimation

To judge client models without access to their data, the server keeps a small validation set $\mathcal{V}$ of exactly 50 images: 10 per grade, sampled at random (seed 42) from the DDR test split, which is separate from all client data. Because $\mathcal{V}$ is balanced, every grade counts equally in its score. In round t, after receiving the locally updated weights $\tilde{w}_k^t$, the server computes

$$\kappa_k^t = \text{QWK}\big(\{y_i\}_{x_i \in \mathcal{V}},\, \{\hat{y}_k^t(x_i)\}_{x_i \in \mathcal{V}}\big) \tag{3}$$

E. Adaptive Cosine Schedule $\beta(t)$

In early rounds, the client models' predictions on $\mathcal{V}$ vary widely, and each κ_k^t is estimated from only 50 images. The aggregation weights are therefore moved gradually from image counts towards quality scores by a half-cosine schedule:

$$\beta(t) = \beta_0 \cos\left(\frac{\pi(t-1)}{2(T-1)}\right) \tag{4}$$

where $\beta_0 = 0.5$ and $T = 10$. In the first round, image counts and quality scores count equally ($\beta(1) = 0.5$); in the last round, $\beta(T) = 0$ and only the quality scores are used.

F. QWK-Weighted Aggregation

The server-set scores are first floored at $\kappa_{\min}$ and turned into quality weights with a temperature-scaled softmax:

$$q_k^t = \frac{\exp\big(\max(\kappa_k^t, \kappa_{\min})/\tau\big)}{\sum_{j=1}^{K} \exp\big(\max(\kappa_j^t, \kappa_{\min})/\tau\big)} \tag{5}$$

where $\kappa_{\min} = 0.1$ and $\tau = 3$. Each client's aggregation weight blends this quality weight with the client's share of the images:

$$\omega_k^t = \beta(t)\, \frac{n_k}{\sum_{j=1}^{K} n_j} + \big(1 - \beta(t)\big)\, q_k^t \tag{6}$$

and the global model is updated as $w^{t+1} = \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$. A minimum weight of $\omega_{\min} = 0.03$ per client, followed by renormalisation, is also enforced as a safeguard; with $\beta_0 = 0.5$ and $K = 5$ it never takes effect, because every client already

receives at least $(1-\beta_0)\,q_{\min} \approx 0.08$. The full procedure is given in Algorithm 1.

The scheme is deliberately conservative. No client is ever excluded, because a single 50-image estimate of κ_k^t is too noisy to justify discarding a hospital's update. With $\tau = 3$ and $\kappa \in [0.1, 1]$, the quality weights q_k^t differ from one another by at most a factor of $e^{0.9/3} \approx 1.35$, so they are close to equal. As $\beta(t)$ decreases, the aggregation therefore moves from image-count weights towards near-equal client weights, with a mild preference for clients that score higher on $\mathcal{V}$. Small clients thus gain influence in later rounds; the uniform-aggregation ablation ($\omega_k^t = 1/K$, Sec. IV-D) separates this effect from that of the QWK scores.

Algorithm 1 OrdinalFed Training Pipeline

Require: Clients K, rounds T, local epochs E, trade-off λ, initial schedule β_0, server set $\mathcal{V}$, initial weights w^0.

for $t = 1$ **to** T **do**
  Broadcast global model w^t to all clients $k \in \{1, \dots, K\}$
  for each client $k \in \{1, \dots, K\}$ **in parallel do**
    $\tilde{w}_k^t \leftarrow \textsc{LocalTrain}(w^t, \mathcal{D}_k, E, \lambda)$ *(Eq. 1)*
    Evaluate validation score κ_k^t on $\mathcal{V}$ *(Eq. 3)*
  end for
  Compute annealing factor $\beta(t)$ *(Eq. 4)*
  Compute aggregation weights ω_k^t for all k *(Eqs. 5–6)*
  $w^{t+1} \leftarrow \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$
end for
return Final model w^T

IV. Experimental Setup

A. Dataset and Preprocessing

All frameworks are evaluated on the **DDR benchmark dataset** [2], which contains 13,673 fundus photographs, 12,522 of them gradable. Excluding the ungradable images leaves 6,260 training, 2,503 validation, and 3,759 test images. All images are resized to 224×224 pixels. During local training, random horizontal flips, rotations of up to $\pm 15°$, and mild colour jitter are applied to reduce overfitting. The training set displays heavy natural class imbalance: Grade 0 (50.0%), Grade 1 (5.0%), Grade 2 (35.8%), Grade 3 (1.9%), and Grade 4 (7.3%).

B. Modeling Clinical Heterogeneity

$K = 5$ clients are instantiated under three statistical scenarios, shown in Fig. 1. In the figure, each bar represents one client: its segments give the share of each DR grade among that client's images, shaded from light (Grade 0) to dark (Grade 4), and the number under each client label is the client's total image count.

- **IID Partition:** Uniform random assignment (1,252 samples per client). This serves as the homogeneous upper bound for FL.
- **Dirichlet Non-IID ($\alpha = 0.1$):** Pathological skew generated via a Dirichlet distribution $p_k \sim \mathrm{Dir}(\alpha)$. An $\alpha = 0.1$ forces each client to draw samples almost exclusively from a single class. Consequently, client sizes range from 32 to 2,087 samples; Client 0 contains solely 32 images (100% Grade 0), while Client 4 holds 2,087 images dominated (98%) by Grade 2.
- **Real-Site Clinical Partition:** Data are distributed according to fixed grade profiles that model five clinical archetypes, yielding: Site 0 (general screening clinic; 40% of images, 84% Grade 0), Site 1 (rural primary care; 22%, mainly Grades 0 and 2, with the largest Grade 1 share at 11%), Site 2 (specialist referral eye hospital; 10%, the largest Grade 3–4 share at 38%), Site 3 (diabetic retinopathy clinic; 19%, 67% Grade 2), and Site 4 (urban hospital; 9%, mixed grades).

C. Network Architecture and Baselines

All federated experiments use an EfficientNet-B3 backbone, chosen for its balance between parameter efficiency and feature extraction capability at 224×224 resolution. To simulate real-world deployment where pre-trained weights are adapted, all FL runs are warm-started from a centralised model trained for 30 epochs (best validation QWK = 0.834, reached at epoch 23). OrdinalFed is benchmarked against: (1) **Centralised Baseline**; (2) **FedAvg** [3]; (3) **FedProx** [4] with proximal coefficient $\mu = 0.01$; and (4) **AccWt**, an accuracy-weighted aggregation baseline in the spirit of server-side validation scoring [9]. AccWt uses the same schedule and softmax weighting as OrdinalFed (Eqs. 4–6) but scores each client by plain accuracy on $\mathcal{V}$ instead of QWK, and its clients train with class-weighted cross-entropy. A fifth configuration, **OrdinalFed + Prox**, adds FedProx's proximal term ($\mu = 0.01$) to the OrdinalFed clients, to test whether the server-side and client-side remedies are complementary. All methods execute $T = 10$ communication rounds with $E = 5$ local epochs, batch size 64, Adam optimiser ($\eta = 10^{-4}$), and the same class-imbalance weights $\mathbf{c}$.

D. Ablations and Evaluation Protocol

To isolate the contribution of each component, three ablations of OrdinalFed are evaluated: (i) OrdinalLoss with sample-count aggregation; (ii) class-weighted cross-entropy with QWK-weighted aggregation; and (iii) class-weighted cross-entropy with uniform aggregation ($\omega_k^t = 1/K$). In addition, three settings of OrdinalFed are tested under $\alpha = 0.1$, each changing one value of the default ($\lambda = 0.5$, $\beta_0 = 0.5$, $\tau = 3$): a weaker ordinal term ($\lambda = 0.1$), quality weights from the first round ($\beta_0 = 0$), and sharper quality weights ($\tau = 0.5$). These settings were fixed before the runs and are compared on mean validation QWK only.

Every configuration is repeated with three random seeds, which control data order and augmentation, and results are reported as mean $\pm$ standard deviation. After each round, the global model is evaluated on the DDR validation split and on the DDR test split excluding the 50 images of $\mathcal{V}$ (3,709 images). For each run, the reported model is the one from the round with the highest validation QWK. The test split is never used for training, aggregation, or model selection, and

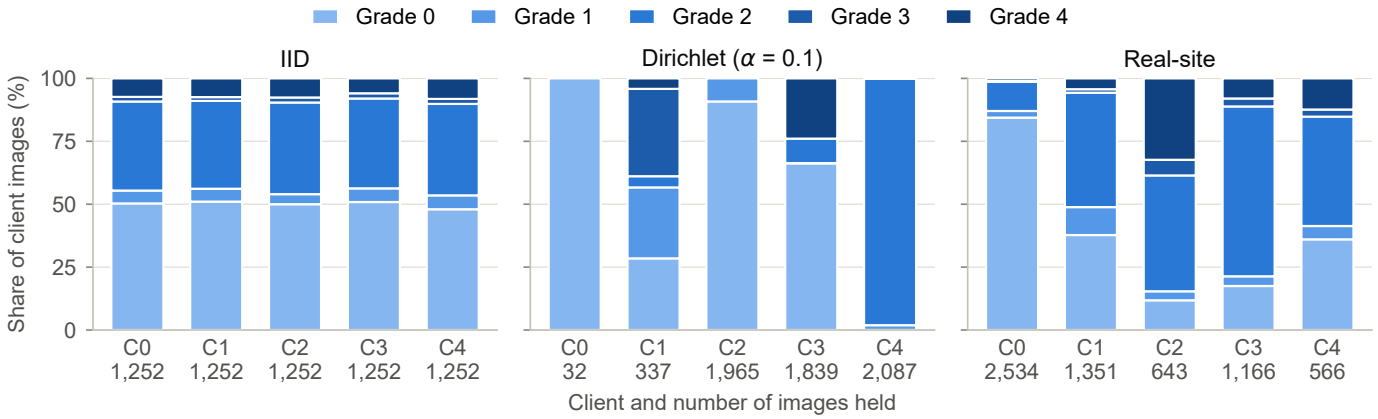


Fig. 1: Per-client grade distributions under the IID, Dirichlet ($\alpha = 0.1$), and real-site partitions. Under $\alpha = 0.1$, client sizes range from 32 images (Client 0, 100% Grade 0) to 2,087 images (Client 4, 98% Grade 2). The real-site partition models five clinical archetypes, from screening clinics to specialist referral centres.

therefore provides an independent estimate of performance. Stability is summarised by the mean, worst and final test QWK over the 10 rounds.

V. Results and Discussion

A. Overall Performance

Table I reports the test QWK of each method at its best validation round, averaged over three seeds. The federated methods reach between 0.640 and 0.681, against 0.682 for the centralised model. On the IID and real-site partitions, the methods differ by at most 0.01, which is within the seed-to-seed variation. Under $\alpha = 0.1$, FedProx scores highest (0.664) and OrdinalFed lowest (0.640). Because the reported round is chosen on the validation split, this table hides what happens in the other rounds: for FedAvg under $\alpha = 0.1$, the best validation round is always the first one, before its collapse.

B. Stability Under Extreme Skew

Fig. 2 plots the test QWK of the global model after every round, with one panel per partition. Lines show the mean over three seeds, shaded bands ± one standard deviation, and the dashed line the test QWK of the centralised model. On the IID and real-site partitions, all three methods stay close to the centralised line. Under $\alpha = 0.1$, FedAvg collapses from round 5 onwards, FedProx stays flat, and OrdinalFed avoids the collapse but oscillates from round to round.

Table II summarises the ten rounds under $\alpha = 0.1$. Compared with FedAvg, OrdinalFed raises the worst round from 0.378 to 0.514 and the mean over rounds from 0.535 to 0.604, without any change at the hospitals. FedProx, which regularises local training instead, is the most stable method (worst round 0.641). Adding FedProx's proximal term to OrdinalFed brings it close to FedProx (worst round 0.627) and gives the best final round (0.664), but it is not more stable than FedProx alone.

TABLE I: Test QWK at the Best Validation Round (Mean ± Std over Three Seeds). Bold: Best Federated Method per Column.

Method	IID	Dirichlet ($\alpha = 0.1$)	Real-Site
Centralised (reference)	0.682	0.682	0.682
FedAvg [3]	0.677 ±.008	0.647 ±.010	0.673 ±.004
FedProx [4]	**0.681** ±.014	**0.664** ±.015	0.671 ±.005
AccWt	–	0.651 ±.011	**0.674** ±.009
OrdinalFed	0.679 ±.006	0.640 ±.007	0.666 ±.002
OrdinalFed + Prox	–	0.659 ±.012	0.672 ±.007

TABLE II: Test QWK Across the 10 Rounds Under Dirichlet Skew ($\alpha = 0.1$): Mean over Rounds, Worst Round and Final Round. Bold: Best per Column.

Method	Mean	Worst	Final
FedAvg	0.535 ±.004	0.378 ±.035	0.475 ±.026
FedProx	**0.659** ±.003	**0.641** ±.008	0.650 ±.006
AccWt	0.627 ±.011	0.571 ±.017	0.657 ±.018
OrdinalFed	0.604 ±.005	0.514 ±.027	0.652 ±.016
OrdinalFed + Prox	0.651 ±.003	0.627 ±.005	**0.664** ±.013

C. What Makes the Aggregation Stable?

Table III separates the two components. OrdinalLoss alone does not prevent the collapse: with image-count weights, its worst round (0.380) matches FedAvg's. QWK-weighted aggregation with cross-entropy raises the worst round to 0.586, but giving every client the same weight raises it further, to 0.630. Scoring clients by accuracy instead of QWK (AccWt, Table II) gives nearly the same result as scoring them by QWK (worst round 0.571 versus 0.586).

The logged aggregation weights explain this. Under $\alpha = 0.1$, the two smallest clients hold 5.9% of the images and therefore 5.9% of FedAvg's weight. Because the quality weights of Eq. (5) are close to equal, OrdinalFed raises their combined weight from 23.5% in round 1 to 40.8% in round 10. The QWK scores themselves carry little information at this scale:

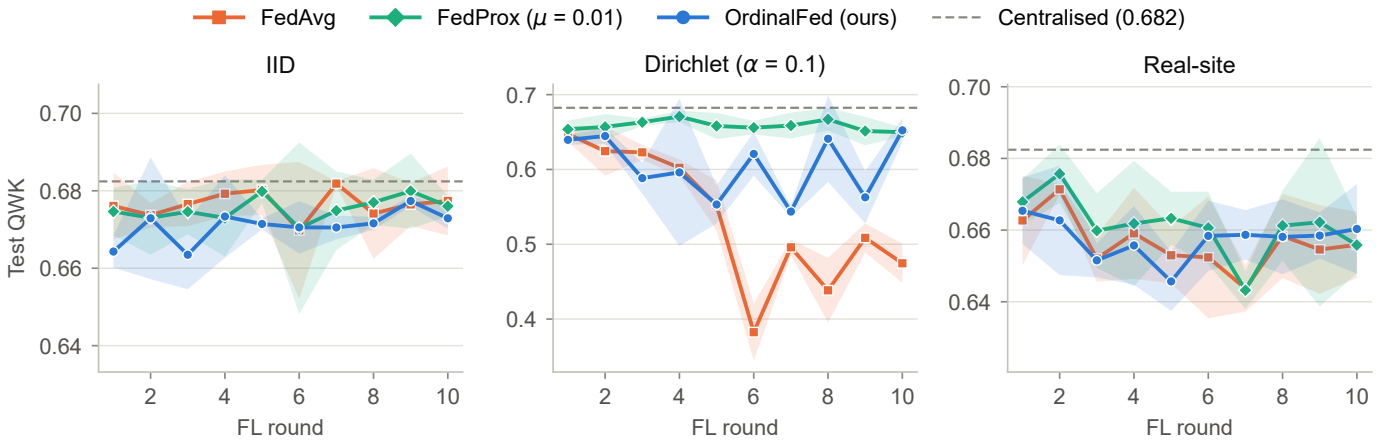


Fig. 2: Test QWK of the global model after each of the 10 rounds for the IID, Dirichlet ($\alpha = 0.1$) and real-site partitions (mean $\pm$ std over three seeds). Under $\alpha = 0.1$, FedAvg collapses, OrdinalFed avoids the collapse but oscillates, and FedProx remains stable.

a client's score on the 50-image set $\mathcal{V}$ changes by 0.13 on average from one round to the next, and in round 10 the largest quality weight is only 1.19 times the smallest. The stability therefore comes from rebalancing weight towards small clients, not from the QWK scores, while OrdinalLoss slightly lowers performance in both comparisons.

The lower block of Table III tests three settings of OrdinalFed, each fixed before the runs and compared on mean validation QWK. All three improve on the default. Sharper quality weights ($\tau = 0.5$) score best (0.795 versus 0.761), followed closely by starting from quality weights ($\beta_0 = 0$, 0.793), but both remain below equal weighting (0.810) and FedProx (0.808).

D. Class-Wise Performance

Table IV gives the per-grade recall and F1 on the test split. All methods detect Grade 0 reliably (recall ≥ 0.95) but almost never Grade 1 (recall ≤ 0.05), which makes up only 5% of the training set. Grade 2 recall stays near 0.4 because many test Grade 2 images are predicted as Grade 0 (Sec. V-F). OrdinalFed has the highest Grade 3 recall in both partitions (0.36 versus 0.27–0.31 under $\alpha = 0.1$, and 0.40 versus 0.35–0.38 on the real-site partition), a modest gain on the rarest severe grade. For Grade 4, the three methods are within 0.05 of each other.

E. Communication and Computational Overhead

OrdinalFed exchanges the same model weights as FedAvg, so it adds no communication. The server evaluates each of the K client models on the 50 images of $\mathcal{V}$ once per round. This took a median of 1.2 s per round for $K = 5$ on an Intel Arc B580 GPU, about 0.6% of the 195 s needed for a full round.

F. Limitations

Grade 1 recall stays at or below 0.05 for every method (Table IV). All models also score lower on the DDR test split than on the validation split: the centralised model drops from a QWK of 0.834 to 0.682, and the federated models drop by a similar amount. Most of this drop comes from Grade 2 images predicted as Grade 0 (about 45% of test Grade 2 images, against about 12% on the validation split), and the test split contains a different mix of camera resolutions. Validation QWK therefore overestimates performance on unseen images, which is why all main results are reported on the test split. The 50-image server set $\mathcal{V}$ is too small to rank clients reliably, which limits the value of QWK-based weighting. Finally, all results come from a single dataset at 224×224 resolution

TABLE III: Contribution of Each Component and Setting Under $\alpha = 0.1$. Mean: QWK Averaged over the 10 Rounds; Worst: Lowest Round. CE: Class-Weighted Cross-Entropy. Default: $\lambda = 0.5$, $\beta_0 = 0.5$, $\tau = 3$.

Configuration	Mean val	Mean test	Worst test
CE + image-count weights (FedAvg)	0.670 $\pm$.011	0.535 $\pm$.004	0.378 $\pm$.035
OrdinalLoss + image-count weights	0.659 $\pm$.015	0.522 $\pm$.011	0.380 $\pm$.018
CE + QWK weights	0.786 $\pm$.006	0.633 $\pm$.010	0.586 $\pm$.006
CE + equal weights	**0.810** $\pm$.007	**0.658** $\pm$.006	**0.630** $\pm$.021
OrdinalFed (default)	0.761 $\pm$.003	0.604 $\pm$.005	0.514 $\pm$.027
with $\lambda = 0.1$	0.776 $\pm$.007	0.622 $\pm$.006	0.571 $\pm$.019
with $\beta_0 = 0$	0.793 $\pm$.005	0.631 $\pm$.009	0.571 $\pm$.011
with $\tau = 0.5$	0.795 $\pm$.006	0.637 $\pm$.005	0.592 $\pm$.009

TABLE IV: Per-Grade Recall / F1 on the Test Split at the Best Validation Round (Mean over Three Seeds).

Method	G0	G1	G2	G3	G4
Dirichlet ($\alpha = 0.1$)					
FedAvg	.96/.80	.05/.07	.41/.52	.27/.32	.66/.73
FedProx	.95/.81	.05/.06	.43/.54	.31/.34	.69/.74
OrdinalFed	.97/.80	.04/.07	.39/.50	.36/.38	.64/.73
Real-site					
FedAvg	.98/.81	.04/.06	.44/.55	.35/.37	.66/.73
FedProx	.98/.81	.04/.06	.43/.54	.38/.38	.65/.73
OrdinalFed	.97/.81	.04/.06	.43/.54	.40/.38	.64/.73

with five clients; larger server sets, higher resolution and more clients are left for future work.

VI. Conclusion

OrdinalFed, a federated framework for multi-grade Diabetic Retinopathy screening that combines a distance-aware OrdinalLoss with QWK-weighted aggregation, has been evaluated on the DDR benchmark with three seeds and a held-out test split. Under extreme label skew, its server-side weighting prevents most of FedAvg's collapse (worst-round test QWK 0.514 versus 0.378) at no communication cost. The ablations show that this benefit comes from giving small hospitals more weight: equal client weights perform better, a 50-image server set is too small for the QWK scores to rank clients, and the OrdinalLoss does not help in the federated setting. FedProx remains the most stable method, and combining it with OrdinalFed comes close to, but does not exceed, FedProx alone. On realistic multi-site and IID partitions, all methods perform alike. Future work will test larger server sets, higher image resolution and cross-dataset validation on APTOS 2019, and will explore formal differential privacy guarantees.

References

[1] International Diabetes Federation, *IDF Diabetes Atlas*, 10th ed. Brussels, Belgium: IDF, 2021. [Online]. Available: https://diabetesatlas.org (accessed Oct. 7, 2026).

[2] T. Li, Y. Gao, K. Wang, S. Guo, H. Liu, and H. Kang, "Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening," *Inf. Sci.*, vol. 501, pp. 511–522, Oct. 2019, doi: 10.1016/j.ins.2019.06.011.

[3] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist. (AISTATS)*, vol. 54, 2017, pp. 1273–1282, doi: 10.48550/arXiv.1602.05629.

[4] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst. (MLSys)*, vol. 2, 2020, pp. 429–450, doi: 10.48550/arXiv.1812.06127.

[5] S. Toledo-Cortés, D. H. Useche, H. Müller, and F. A. González, "Grading diabetic retinopathy and prostate cancer diagnostic images with deep quantum ordinal regression," *Comput. Biol. Med.*, vol. 145, Art. no. 105472, Jun. 2022, doi: 10.1016/j.compbiomed.2022.105472.

[6] G. Polat, Ü. M. Çağlar, and A. Temizel, "Class distance weighted cross entropy loss for classification of disease severity," 2024, *arXiv:2412.01246*, doi: 10.48550/arXiv.2412.01246.

[7] D. Bhulakshmi and D. S. Rajput, "FedDL: Personalized federated deep learning for enhanced detection and classification of diabetic retinopathy," *PeerJ Comput. Sci.*, vol. 10, Art. no. e2508, Dec. 2024, doi: 10.7717/peerj-cs.2508.

[8] G. Ciaramella *et al.*, "A privacy-preserving and explainable method for diabetic retinopathy classification and localization," *IEEE Access*, vol. 14, pp. 49830–49850, 2026, doi: 10.1109/ACCESS.2026.3678201.

[9] I. Bejenar, L. Ferariu, C. Pascal, and C.-F. Caruntu, "Aggregation methods based on quality model assessment for federated learning applications: Overview and comparative analysis," *Mathematics*, vol. 11, no. 22, Art. no. 4610, Nov. 2023, doi: 10.3390/math11224610.

[10] Z. Li, T. Lin, X. Shang, and C. Wu, "Revisiting weighted aggregation in federated learning with neural networks," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 19767–19788, doi: 10.48550/arXiv.2302.10911.

[11] X. Li, Y. Gao, Y. Deng, and X. Jiang, "Federated learning with adaptive aggregation weights for non-IID data in edge networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 11, no. 5, pp. 3425–3439, Oct. 2025, doi: 10.1109/TCCN.2025.3534248.

[12] A. Mora, A. Bujari, and P. Bellavista, "Enhancing generalization in federated learning with heterogeneous data: A comparative literature review," *Future Gener. Comput. Syst.*, vol. 157, pp. 1–15, Aug. 2024, doi: 10.1016/j.future.2024.03.027.